数理分析方法

连续函数与半连续性

课件编号：107-ZSM-PPT　原始 PPT：43 页

逐页保留原始 PPT 画面 + 中文零基础自学讲解 + 证明专题

阅读说明：本讲义每一页原课件均以原始 PDF 页面完整嵌入；其后的中文解读旨在帮助初学者识别定义、假设、论证步骤。章末为从零展开的核心证明与复习提示。讲义属于学习辅助材料，不代替原教师课件。参考：Gérard Debreu (1959), *Theory of Value*。

目录

1 学前导读

这一讲学什么

连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

掌握三种语言：ε–δ、序列保持极限、开（闭）集逆像。考试中根据已知条件选择最省力的一种。

证明题通用方法（先读本页）

严格证明的四个必要动作

第一：写出假设。比如“K 非空、闭、有界”，不能只写“K 很好”。

第二：逐字展开定义。写出 $\forall$（任意）、$\exists$（存在）、$\Rightarrow$（推出）的顺序。

第三：解释每一次定理调用。从有界得到收敛子列还缺 Bolzano–Weierstrass；得到极限在原集合还需要闭性。

第四：检查结论是否恰好是题目所求。“有极限”不等于“极限属于集合”，“函数有界”不等于“达到最大值”。

最常用的三种模板

集合相等：任取 x，证明 $x \in A \Rightarrow x \in B$，再证明 $x \in B \Rightarrow x \in A$。

极限：给定任意 $\varepsilon > 0$，找出一个 N（或 δ）并验证最终不等式。

存在性：常用反证、上确界构造、嵌套区间、连续函数极值、固定点定理；对每个方法要明确其全部前提。

2 逐页课件与中文讲解

2.1 第 1 页｜本页课件

Theory of Value

An Axiomatic Analysis of Economic Equilibrium

Gerard Debreu (1959)

Professor Shunming Zhang

School of Finance
Renmin University of China

September 26, 2022

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

结合原文理解：掌握三种语言：ε–δ、序列保持极限、开（闭）集逆像。考试中根据已知条件选择最省力的一种。

读课件时的动作：本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.2 第 2 页 | Chapter 1: Mathematics

Chapter 1: Mathematics

01. Introduction
02. Sets
03. Functions and Correspondences
04. Preorderings
05. Real Numbers
06. Limits in $\mathbb{R}^N$
07. Continuous Functions
08. Continuous Correspondences
09. Vectors in $\mathbb{R}^N$
10. Fixed Points

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点： 反身性是 xRx；传递性是 xRy 且 yRz　xRz。完全性与反对称性不包含在预序定义内。

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2.3 第 3 页｜Continuous Functions

Continuous Functions

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, (x^q) a sequence of points of S, and (y^q) a sequence of points of T.

这一页怎么读

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2.4 第 4 页｜Continuous Functions

Continuous Functions

Let f be a function from S to T, and consider a point $x^0 \in S$. The function f is continuous at the point x^0 if:

$$\text{“}x^q \to x^0, y^0 = f\left(x^0\right), y^q = f\left(x^q\right)\text{”} \text{ implies “}y^q \to y^0\text{”}$$

In other words. f is continuous at x^0 if the image of any sequence tending to x^0 is a sequence tending to the image of x^0.

Example: Let f be the real-valued function of a real variable (i.e., a function from $\mathbb{R}$ to $\mathbb{R}$) defined by $y = \frac{1}{x}$ if $x \neq 0$, and $y = 0$ if $x = 0$. The function f is continuous at every point of $\mathbb{R}$, with the exception of 0, as its graph readily suggests.

The function f is continuous on S if it is continuous at every point of S.

这一页怎么读

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2.5 第 5 页｜Continuous Functions

Continuous Functions

Let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2 and g be a function from S_2 to T. Define a function h from S_1 to T by $h(x) = g(f(x))$ for every x in S_1. It is immediate that:

(1) If f is continuous at the point x of S_1, and if g is continuous at the point $f(x)$ of S_2, then h is continuous at x.

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2.6 第 6 页｜Continuous Functions

Continuous Functions

For every $n = 1, 2, \cdots, N$, let T_n be a subset of $\mathbb{R}^{L_n}$ and consider the product $T = \prod_{n=1}^{N} T_n$. It is immediate that the projection on T_n is continuous on T.

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读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.7 第 7 页｜Continuous Functions

Continuous Functions

With the notations of the last paragraph, let f_n be a function from S to T_n, and define a function f from S to T by $f(x) = (f_n(x))$ for every x in S, where $(f_n(x))$ denotes the N-tuple $(f_1(x), f_2(x), \cdots, f_N(x))$. It is immediate that:

(2) If every f_n is continuous at the point x of S, then f is continuous at x.

这一页怎么读

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2.8 第 8 页｜Continuous Functions

Continuous Functions

One can prove:

(3) A function f from S to T is continuous on S if and only if the inverse image of every set closed in T is closed in S.

Proposition: A mapping f from a metric space X to a metric space Y. The following statements are equivalent:

(1) f is continuous.

(2) For each open set G in Y, the set $f^{-1}(G)$ is an open set in X.

(3) For each closed set F in Y, the set $f^{-1}(F)$ is an closed set in X.

(4) For every set A in X, $f(\overline{A}) \subseteq \overline{f(A)}$.

(5) For every set B in Y, $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$.

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2.9 第 9 页｜Continuous Functions

Continuous Functions

Proof: The function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|x - x_0\| < \delta \qquad \text{implies} \qquad \|f(x) - f(x_0)\| < \varepsilon,$$

or, the function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in N_\delta(x_0) \qquad \text{implies} \qquad f(x) \in N_\varepsilon(f(x_0)),$$

that is, $f(N_\delta(x_0)) \subseteq N_\varepsilon(f(x_0))$, equivalently, the function f is continuous at the point x_0 if for any neighborhood V of $f(x_0)$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$.

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2.10 第 10 页｜Continuous Functions

Continuous Functions

Proof of (1) $\Longrightarrow$ (2): For each open set G in Y, we want to check that set $f^{-1}(G)$ is an open set in X.
For any $x_0 \in f^{-1}(G)$, $f(x_0) \in G$. For any neighborhood V of $f(x_0)$ with $f(x_0) \in V \subseteq G$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$, then $x_0 \in U \subseteq f^{-1}(V) \subseteq f^{-1}(G)$.
Proof of (2) $\Longrightarrow$ (1): For any neighborhood V of $f(x_0)$, there exists neighborhood V' of $f(x_0)$ such that $f(x_0) \in V' \subseteq V$, then $x_0 \in f^{-1}(V') \subseteq f^{-1}(V)$ and $f^{-1}(V')$ is an open set.
Proof of (2) $\Longrightarrow$ (3): For each closed set F in Y, $G \equiv Y \setminus F$ is open, then
$f^{-1}(G) = f^{-1}(Y \setminus F) = f^{-1}(Y) \setminus f^{-1}(F) = X \setminus f^{-1}(F)$ is open, then the set $f^{-1}(F)$ is an closed set in X.
Proof of (3) $\Longrightarrow$ (2): For each open set G in Y, $F \equiv Y \setminus G$ is closed, then
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对应的完整证明：参见本讲义最后的「closed」证明专题。

2.11 第 11 页｜Continuous Functions

Continuous Functions

Proof of (3) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq \overline{f(A)}$, then $A \subseteq f^{-1}\left(\overline{f(A)}\right)$ and $f^{-1}\left(\overline{f(A)}\right)$ is a closed set in X, hence $\overline{A} \subseteq f^{-1}\left(\overline{f(A)}\right)$. Thus $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.

Proof of (4) $\Longrightarrow$ (3): For each closed set F in Y, define $A = f^{-1}(F)$, then $f(A) = f\left(f^{-1}(F)\right) \subseteq F$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)} \subseteq \overline{F} = F$. Thus $\overline{A} \subseteq f^{-1}(F) = A$, $A = \overline{A}$ is closed, that is, the set $f^{-1}(F)$ is an closed set in X.

Proof of (3) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq f^{-1}\left(\overline{B}\right)$ and $f^{-1}\left(\overline{B}\right)$ is a closed set in X, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (3): For each closed set F in Y, $\overline{f^{-1}(F)} \subseteq f^{-1}\left(\overline{F}\right) = f^{-1}(F)$, then the set $f^{-1}(F)$ is an closed set in X.

Proof of (4) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq X$, then $f\left(\overline{f^{-1}(B)}\right) \subseteq \overline{f\left(f^{-1}(B)\right)} \subseteq \overline{B}$, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq Y$, then $\overline{A} \subseteq \overline{f^{-1}(f(A))} \subseteq f^{-1}\left(\overline{f(A)}\right)$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)}$. □

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2.12 第 12 页｜Continuous Functions

Continuous Functions

In the important particular case where f is a real-valued function on S. i.e., where f is a function from S to $\mathbb{R}$, one can prove:

(3′) A function from S to $\mathbb{R}$ is continuous on S if and only if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S.

Example: (1) Let $\mathbb{R}_+$ be the set of positive reals, and consider the function f from $S = \mathbb{R} \times \mathbb{R}_+$ to $\mathbb{R}$ defined by $y = f(u, v) = \frac{u}{v}$. "The function f is continuous on S" is equivalent to "the inverse image of $(-\infty, y]$, i.e, the set $\{(u, v) \in S \mid u - yv \leqq 0\}$, is closed in S, and similarly for $[y, \infty)$; the second assertion is readily suggested by a drawing in $\mathbb{R}^2$, and easily proved. (2) The example of 1.7.b does not satisfy the continuity criterion of (3′), as the graph shows.

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2.13 第 13 页｜本页课件

Continuous Functions

Proof: If A function from S to $\mathbb{R}$ is continuous on S, as every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in $\mathbb{R}$, then the inverse image is also closed in S.

Conversely, if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S. Assume the function is $f : S \to \mathbb{R}$. So for any open interval of $\mathbb{R}$, the form (a, b), (a, ∞), $(-\infty, a)$, we have

$$f^{-1}(-\infty, a) = f^{-1}(\mathbb{R}) \setminus f^{-1}[a, \infty)$$
$$f^{-1}(a, \infty) = f^{-1}(\mathbb{R}) \setminus f^{-1}(-\infty, a]$$
$$f^{-1}(a, b) = f^{-1}(-\infty, b) \setminus f^{-1}(-\infty, a]$$

are all open in S.

We also have following lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals. So for any open set $O \subseteq \mathbb{R}$, there exists countable open interval $\{I_n; n \in \mathbb{N}\}$ such that $O = \bigcup_{n=1}^{\infty} I_n$. So

$$f^{-1}(O) = \bigcup_{n=1}^{\infty} f^{-1}(I_n)$$

are also open in S. Hence f is continuous. □

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2.14 第 14 页｜Continuous Functions

Continuous Functions

Lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals.

Proof: Let G be a non-empty open set in $\mathbb{R}$. Write $a \sim b$ if the closed interval $[a, b]$ if $a < b$ or $[b, a]$ if $b < a$, lies in G. This is an equivalence relation, in particular $a \sim b$ since $\{a\}$ is itself a closed interval. G is therefore the union of disjoint equivalence classes. Let $C(a)$ be the equivalence class containing a. Let $G = \bigcup_{a \in I} C(a)$ where $C(a) \neq C(a')$ if $a \neq a'$. $C(a)$ is clearly an interval. Also $C(a)$ is open, for if $k \in C(a)$, then $(k - \varepsilon, k + \varepsilon) \subseteq G$ for sufficiently small ε. But then $(k - \varepsilon, k + \varepsilon) \subseteq C(a)$; so G is the union of disjoint intervals. For each $C(a)$ choose a rational q_a, this map $I \to \mathbb{Q} : G(a) \longmapsto q_a$ is injective, so I is countable. □

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本页重点：闭区间套的核心是：左端点递增、有上界，取上确界后必须证明它落在每一个区间内。

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2.15 第 15 页｜ Continuous FunctionsOne can prove:

Continuous Functions

One can prove:

(4) Let f be a function from S to T. If f is continuous on S, and if S is compact, then $f(S)$ is compact.

Proof: Let $\bigcup_{\gamma\in\Gamma} U_\gamma$ be an open covering of $f(S)$ on T. Then the sets $f^{-1}(U_\gamma)$ are open and $\left\{f^{-1}(U_\gamma);\gamma\in\Gamma\right\}$ form an open covering of S since we have show

$$f^{-1}\left(\bigcup_{\gamma\in\Gamma} U_\gamma\right)=\bigcup_{\gamma\in\Gamma} f^{-1}(U_\gamma)$$

Since S is compact, there exists a finite subset $\Gamma'\subseteq\Gamma$ such that $\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)$ still forms a covering of S, then

$$f(S)=f\left(\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)\right)=f\left(f^{-1}\left(\bigcup_{\gamma\in\Gamma'} U_\gamma\right)\right)\subseteq\bigcup_{\gamma\in\Gamma'} U_\gamma$$

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本页重点： 在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

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2.16 第 16 页｜Continuous Functions

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately:

(4′) (Weierstrass' Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is non-empty compact, then $f(S)$ has a maximum and a minimum.

Proof: $f(S)$ is a compact subset of $\mathbb{R}$, hence closed and bounded. Now, any bounded set $A \subseteq \mathbb{R}$ has a least upper bound $\sup A$ and a greatest lower bound $\inf A$, and these two points belong to the closure A. But applying this to $A = f(S)$, which is closed, we conclude that $\sup f(S)$ and $\inf f(S)$ belong to $f(S)$ itself, which is exactly what is being claimed. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：上确界不是一定被取到的最大值。验证 s=sup A：所有 a ∈ A 满足 a≤s，并且任意 ε>0 总有 a ∈ A 满足 s−ε<a。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.17 第 17 页｜Continuous Functions

Continuous Functions

"$f(S)$ has a maximum" also means (1.5.g) "there is an x^M in S such that for all $x \in S$ one has $f(x) \leqq f(x^M)$." Such an element is called a maximizer of f. "$f(S)$ has a minimum" also means (1.5.g) "there is an x^m in S such that for all $x \in S$ one has $f(x^m) \leqq f(x)$." Such an element is called a minimizer of f. The maximum (resp. minimum) of the set $f(S)$ is also called the maximum (resp. minimum) of the function f on S.

Example: Let f be the function defined in the example of 1.7.b, and I an interval to which the variable x in restricted. One sees on the graph of f how, when I is not bounded, or not closed, or owns 0 and a different point, f has no maximizer or no minimizer.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

结合原文理解：掌握三种语言：ε–δ、序列保持极限、开（闭）集逆像。考试中根据已知条件选择最省力的一种。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.18 第 18 页｜Continuous Functions

Continuous Functions

It is easy to prove (using (3)),

(5) Let f be a function from S to T. If f is continuous on S, and if S is connected, then $f(S)$ is connected.

Proof: If $f(S)$ is not connected in Y, then we can find two non-empty and disjoint open subsets V_1 and V_2 of $f(X)$ such that $V_1 \cup V_2 = f(S)$. But then, $f^{-1}(V_1)$ and $f^{-1}(V_2)$ are nonempty open subsets of S. Moreover, these sets are disjoint, and we have $f^{-1}(V_1) \cup f^{-1}(V_2) = f^{-1}(V_1 \cup V_2) = S$. Thus S is not connected. Contradiction! $\square$

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.19 第 19 页｜本页课件

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately (see (8) of 1.6.u):

(5′) (Bolzano's Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is connected, then $f(S)$ is an interval.

Thus, if x^1, x^2 are two points of S, and y is a real number such that $f(x^1) \leqq y \leqq f(x^2)$, then there is an $x \in S$ such that $f(x) = y$. In other words, f takes on all values between $f(x^1)$ and $f(x^2)$. This result is often applied to the case where S is an interval of $\mathbb{R}$.

The Intermediate Value Theorem. Let X be a connected metric space and $f : X \to \mathbb{R}$ a continuous function. If $f(x_1) \leqslant \alpha \leqslant f(x_2)$ for some $x_1 \in X$ and $x_2 \in X$, then there exists a $x \in X$ such that $f(x) = \alpha$.

Proof. $f(X)$ is connected in $\mathbb{R}$. Hence $f(X)$ must be an interval. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：介值定理常设 A=x:f(x)≤0，令 c=sup A，再用连续性排除 f(c)≠0。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

对应的完整证明：参见本讲义最后的「ivt」证明专题。

2.20 第 20 页｜本页课件

Notions of Continuity for Functions

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. When it is clear to which space a pair of points belongs, I may drop the subscripts on d_X and d_Y and simply write d for the metric. I will also use the notion $B_\varepsilon(x)$ to denote the open ball of radius ε centered at x when the metric is obvious.

Recall that the Euclidean metric on $\mathbb{R}^N$ (or l^2) is given by

$$d_2\left(x^1, x^2\right) = \sqrt{\sum_{n=1}^{N}\left(x_n^1 - x_n^2\right)^2}$$

Other useful metrics on $\mathbb{R}^N$ include the max (or l^∞) metric

$$d_\infty\left(x^1, x^2\right) = \max\left\{\left|x_n^1 - x_n^2\right| : 1 \leqslant n \leqslant N\right\}$$

and the taxicab (or l^1) metric

$$d_1\left(x^1, x^2\right) = \sum_{n=1}^{N}\left|x_n^1 - x_n^2\right|$$

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

结合原文理解： 掌握三种语言：ε–δ、序列保持极限、开（闭）集逆像。考试中根据已知条件选择最省力的一种。

读课件时的动作： 本页用于承接前后概念。阅读时要回答两个问题：这一页新增了什么条件？它将用于下一页的哪一个结论？

2.21 第 21 页｜Notions of Continuity for Functions

Notions of Continuity for Functions

Definition 1 (Continuity using neighborhoods). A function $f : X \to Y$ is continuous at x if for every neighborhood V of $f(x)$, its inverse image $f^{-1}(V)$ is a neighborhood of x. That is, for every open set V with $f(x) \in V$, there exists an open set U with $x \in U$ such that

$$z \in U \Longrightarrow f(z) \in V,$$

i.e., $f(U) \subseteq V$.

Note that this definition is topological, that is, it uses only the notion of neighborhoods and does not mention the metrics. The next definition makes explicit use of the metrics.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：“像”先选 x 再映到 y；“逆像”先限定 y 的目标集合，再找所有满足 f(x)　B 的 x。

读课件时的动作：本页以定义为主。建议将英文谓词翻译成“任意…存在…”的中文句子，再试着写一个满足定义和一个不满足定义的例子。

2.22 第 22 页｜Notions of Continuity for Functions

Notions of Continuity for Functions

Definition 2 (Continuity using metrics). A function $f : X \to Y$ is continuous at x if for every $\varepsilon > 0$ there exists some $\delta > 0$ such that

$$d_X(x, z) < \delta \Longrightarrow d_Y(f(x), f(z)) < \varepsilon.$$

Definition 3 (Continuity using sequences). A function $f : X \to Y$ is continuous at x if whenever $x_n \to x$, then $f(x_n) \to f(x)$.

Proposition 1. A function f is continuous at x in the topological sense of Definition 1 if and only if it is continuous at x in the metric sense of Definition 2 if and only if it is continuous at x in the sequential sense of Definition 3.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

结合原文理解：掌握三种语言：ε–δ、序列保持极限、开（闭）集逆像。考试中根据已知条件选择最省力的一种。

读课件时的动作：本页是定理或命题。先明确量词、集合范围、连续性及闭/紧/凸假设；再检查结论是“存在”“唯一”还是“对于任意”。定理本身不能替代证明。

2.23 第 23 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

The ideas of continuity introduced above apply in particular to numerical functions and we say that a numerical function f, defined on a topological space X, is continuous at a point x_0 if it is a mapping of X into $\mathbb{R}$ and, as such, is continuous at x_0; in other words if, to each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.24 第 24 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

In certain situations this idea turns out to be too restrictive and we are forced to consider the concepts of lower and upper semi-continuity, as given in the following definitions.

A numerical function f defined on X is said to be lower semi-continuous (l.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) > f(x_0) - \varepsilon.$$

A numerical function f defined on X is said to be upper semi-continuous (u.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) < f(x_0) + \varepsilon.$$

We note that f is continuous at x_0 if and only if it is both lower and upper semi-continuous at that point.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

读课件时的动作：本页有例子或构造。自己换一个简单数值例子，算出具体集合或函数值；再思考一旦撤掉某条假设，结论是否还成立。

2.25 第 25 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Properties of upper semi-continuous functions can easily be deduced from those of lower semi-continuous functions by replacing $f(x)$ by $-f(x)$; we shall therefore consider only the case of lower semi-continuous functions.

We say that a function f is lower semi-continuous (l.s.c.) (in X) if it is lower semi-continuous at each point of X.

We say that a function f is upper semi-continuous (u.s.c.) (in X) if it is upper semi-continuous at each point of X.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

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2.26 第 26 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open.

Proof. The family $\mathscr{D}^-$ of intervals

$$\{(\lambda, +\infty) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of lower semi-continuity that f is lower semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^-)$. Since $S_\lambda^- = f^{-1}((\lambda, +\infty))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is lower semi-continuous if and only if, for each number λ, the lower contour set $T_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.27 第 27 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open.

Proof. The family $\mathscr{D}^+$ of intervals

$$\{(-\infty, \lambda) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of upper semi-continuity that f is upper semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^+)$. Since $S_\lambda^+ = f^{-1}((-\infty, \lambda))$, the theorem stated is equivalent to this result. □

Corollary 1. A numerical function f is upper semi-continuous if and only if, for each number λ, the upper contour set $T_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.28 第 28 页｜ Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open, i.e. the lower contour set $T_\lambda^- = \mathbf{C}_X S_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)-\epsilon}^- = \{x \in X \mid f(x) > f(x_0) - \epsilon\}$ is an open set and $x \in S_{f(x_0)-\epsilon}^-$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)-\epsilon}^-$, then $f(x) > f(x_0) - \epsilon$ for all $x \in U$. Hence f is lower semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is lower semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$. For any $x_0 \in S_\lambda^-$, $f(x_0) > \lambda$, then let $\epsilon = \dfrac{f(x_0) - \lambda}{2}$, there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $f(x_0) - \epsilon = \dfrac{f(x_0) + \lambda}{2} > \lambda$, $U \subseteq S_\lambda^-$, then S_λ^- is open. □

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点： 开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

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对应的完整证明： 参见本讲义最后的「closed」证明专题。

2.29 第 29 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open, i.e. the upper contour set $T_\lambda^+ = \mathbf{C}_X S_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)+\epsilon}^+ = \{x \in X \mid f(x) < f(x_0) + \epsilon\}$ is an open set and $x \in S_{f(x_0)+\epsilon}^+$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)+\epsilon}^+$, then $f(x) < f(x_0) + \epsilon$ for all $x \in U$. Hence f is upper semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is upper semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$. For any $x_0 \in S_\lambda^+$, $f(x_0) < \lambda$, then let $\epsilon = \dfrac{\lambda - f(x_0)}{2}$, there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $f(x_0) + \epsilon = \dfrac{f(x_0) + \lambda}{2} < \lambda$, $U \subseteq S_\lambda^+$, then S_λ^+ is open. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

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2.30 第 30 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a lower semi-continuous function f attains in K the value $m = \inf_{x \in K} f(x)$ (in other words, $\inf_{x \in K} f(x)$ is a minimum).

Proof. Let μ be a number such that $\mu > m$. Then $S_\mu^- = \{x \in K \mid f(x) \leqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu > m} S_\mu^- \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = m$. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：在有限维欧氏空间，紧致 ⇔ 闭且有界 ⇔ 每个序列有收敛子列且极限仍在集合内。

读课件时的动作：本页出现推导或证明。把每一行改写为“由哪条定义或假设，得到哪一个新结论”。若老师写了“显然”，不要跳过：你至少应能写出一条完整的蕴含式。

2.31 第 31 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a lower semi-continuous function on a compact metric space X. Then f is bounded below and attains its minimum.

Proof. Let $U_\lambda^- = (\lambda, +\infty)$. $\left\{U_\lambda^-; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^-\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^-\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^-\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^-\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^-$$

hence $f(x) \geqq \min\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded below.

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：在有限维欧氏空间，紧致 $\Leftrightarrow$ 闭且有界 $\Leftrightarrow$ 每个序列有收敛子列且极限仍在集合内。

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2.32 第 32 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Let $m = \inf_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is lower semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_0) - \epsilon < f(x_{n_k})$$

let $k \to \infty$, we have

$$f(x_0) - \epsilon \leqq m.$$

As ϵ is arbitrary, we have $f(x_0) \leqq m$, and by definition of m, we have $f(x_0) \geqq m$, hence x_0 is a minimizer. $\square$

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.33 第 33 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a upper semi-continuous function f attains in K the value $M = \sup_{x \in K} f(x)$ (in other words, $\sup_{x \in K} f(x)$ is a maximum).

Proof. Let μ be a number such that $\mu < M$. Then $S_\mu^+ = \{x \in K \mid f(x) \geqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu < M} S_\mu^+ \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = M$. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.34 第 34 页｜ Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a upper semi-continuous function on a compact metric space X. Then f is bounded above and attains its maximum.

Proof. Let $U_\lambda^+ = (-\infty, \lambda)$. $\left\{U_\lambda^+; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^+\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^+\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^+\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^+\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^+$$

hence $f(x) \leqq \max\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded above.

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.35 第 35 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Let $M = \sup_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is upper semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_{n_k}) < f(x_0) + \epsilon$$

let $k \to \infty$, we have

$$M \leqq f(x_0) + \epsilon.$$

As ϵ is arbitrary, we have $M \leqq f(x_0)$, and by definition of M, we have $f(x_0) \leqq M$, hence x_0 is a maximizer. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.36 第 36 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of lower semi-continuous functions, the function g such that $g(x) = \sup_{i \in I} f_i(x)$ is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) > g(x_0) - \varepsilon.$$

Since f_i is lower semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies f_i(x) > f_i(x_0) - \varepsilon > g(x_0) - 2\varepsilon$$

and therefore

$$x \in U(x_0) \implies g(x) \geqq g(x_0) - 2\varepsilon.$$

Hence g is lower semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：下半连续要求：若 y Φ(x)，当自变量序列趋近给定点时，可以从临近的对应值集中挑选一列点逼近原来的目标点（在适当的度量空间设定下）。

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2.37 第 37 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of upper semi-continuous functions, the function g such that $g(x) = \inf_{i \in I} f_i(x)$ is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) < g(x_0) + \varepsilon.$$

Since f_i is upper semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \implies f_i(x) < f_i(x_0) + \varepsilon < g(x_0) + 2\varepsilon$$

and therefore

$$x \in U(x_0) \implies g(x) \leqq g(x_0) + 2\varepsilon.$$

Hence g is upper semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. □

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点： 上半连续对应可用“闭集上逆像为闭集”或序列判别，在紧值等假设下成立。不要把闭图像直接当成普遍等价条件。

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2.38 第 38 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are lower semi-continuous functions then the function $g(x) = \inf_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) > f_n(x_0) - \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) > f_n(x_0) - \varepsilon,$$

whence

$$g(x) = \inf_{n \in I} f_n(x) \geqq \inf_{n \in I} \{f_n(x_0) - \varepsilon\} = g(x_0) - \varepsilon$$

and so g is lower semi-continuous. □

这一页怎么读

概念主线： 连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.39 第 39 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are upper semi-continuous functions then the function $g(x) = \sup_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) < f_n(x_0) + \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) < f_n(x_0) + \varepsilon,$$

whence

$$g(x) = \sup_{n \in I} f_n(x) \leqq \sup_{n \in I} \{f_n(x_0) + \varepsilon\} = g(x_0) + \varepsilon$$

and so g is upper semi-continuous. □

这一页怎么读

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2.40 第 40 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two lower semi-continuous numerical functions, the function $f_1 + f_2$ is lower semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) > a\} = \bigcup_{\lambda} \{x \mid f_1(x) > a - \lambda\} \cap \{x \mid f_2(x) > \lambda\}$$

is a union of open sets and so is open. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

本页重点：开集允许在每一点周围放一个小球；闭集包含所有属于它的收敛序列的极限。

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2.41 第 41 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two upper semi-continuous numerical functions, the function $f_1 + f_2$ is upper semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) < a\} = \bigcup_{\lambda} \{x \mid f_1(x) < a - \lambda\} \cap \{x \mid f_2(x) < \lambda\}$$

is a union of open sets and so is open. □

这一页怎么读

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2.42 第 42 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive lower semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is lower semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) > \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) > \lambda\} \cap \left\{x \mid f_2(x) > \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

这一页怎么读

概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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2.43 第 43 页｜Numerical Functions Defined on a Topological Space

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive upper semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is upper semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) < \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) < \lambda\} \cap \left\{x \mid f_2(x) < \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. □

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概念主线：连续性意味着输入小变化导致输出小变化；紧集上的连续函数具有取极值、保持紧集和连通性等性质。

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3 证明专题：考试应会的标准推导

专题的使用方法

这些专题沿着本课 PPT 的定理顺序补足主要推导；若原课件的命题另有附加假设，以该页原文为主，切勿将证明中的条件删掉。对于高维拓扑不动点定理，只陈述严格的适用条件并说明所需工具，不伪称给出了没有依据的完整高维证明。

3.1 专题：closed

证明｜闭集的序列刻画

若 $F \subset \mathbb{R}^N$ 闭，则其补集开。设 $x_n \in F$ 且 $x_n \to x$；若 $x \notin F$，补集含 x 的某个开球，收敛序列最终进入该球，与 $x_n \in F$ 矛盾。反之，若 F 包含内部任意收敛序列的极限，但补集不开放，则存在 $x \notin F$，使每个 $B(x, 1/n)$ 都与 F 相交。选 $x_n \in F \cap B(x, 1/n)$，得 $x_n \to x \notin F$，矛盾。因此 F 闭。

3.2 专题：connected

证明｜连续映射保持连通性

设 X 连通，$f : X \to Y$ 连续，反设 $f(X)$ 不连通，则存在互不相交、非空且相对开放的 $U, V \subset f(X)$，使 $f(X) = U \cup V$。由连续性 $f^{-1}(U)$、$f^{-1}(V)$ 在 X 中相对开放；它们非空、不相交并覆盖 X，与 X 连通矛盾。故 $f(X)$ 连通。

3.3 专题：evt

证明｜连续函数在非空紧集上取得最大、最小值

设 $K \subset \mathbb{R}^N$ 非空紧致，$f : K \to \mathbb{R}$ 连续。先证有界：若无上界，取 $x_n \in K$ 使 $f(x_n) > n$。紧致性给出收敛子列 $x_{n_k} \to x \in K$，连续性给出 $f(x_{n_k}) \to f(x)$，与 $f(x_{n_k}) > n_k \to \infty$ 矛盾。令 $M = \sup_{x \in K} f(x)$，依上确界性质取 $x_n \in K$ 使 $f(x_n) > M - 1/n$，取收敛子列 $x_{n_k} \to x^* \in K$，连续性推出 $f(x^*) = M$。对 $-f$ 重复论证即得最小值。

3.4 专题：ivt

证明｜介值定理（以零点为例）

设 $f : [a, b] \to \mathbb{R}$ 连续，$f(a) < 0 < f(b)$。令 $A = \{x \in [a, b] : f(x) \le 0\}$，因 $a \in A$，A 非空且有上界 b，设 $c = \sup A$。若 $f(c) < 0$，连续性保证 c 右侧足够近的点仍使 $f < 0$，违背 c 为上界。若 $f(c) > 0$，连续性保证 c 左侧足够近的点都使 $f > 0$，违背 c 是 A 的最小上界（用上确界的 ε 刻画）。因此 $f(c) = 0$。一般介值 v 取 $g(x) = f(x) - v$ 即可。

4 自测清单

1. 你能不看课件写出本讲的核心定义及所有量词吗?
2. 你能逐条列出每一个定理的假设，并解释删去一条为何可能失败吗?
3. 你能在白纸上独立复述本讲证明专题中的每一个箭头吗?
4. 你能结合原课件的反例判断是否把充分条件错写成必要条件吗?